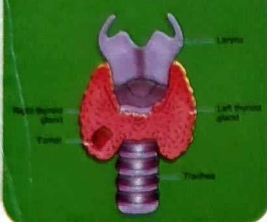


CHAPTER

35



Thyroid Disorders

Learning Objectives

- ⇒ To know the functions of thyroid gland in brief
- ⇒ To have a comprehensive understanding of thyroid diseases
- ⇒ How to perform the physical examination of the patient of thyroid disease
- ⇒ To know about the investigations preferred and their interpretation

Thyroid gland is one of the major endocrine glands which controls all of metabolic processes of the body. This gland is situated in the anterior neck just below the cricoid cartilage, consists of 2 lobes connected by an isthmus. The diseases affecting thyroid gland can range from harmless swelling termed goitre to life threatening cancer. The most common thyroid abnormalities manifested in human beings is due to abnormal production of thyroid hormones. It could be the increased production of hormones termed as hyperthyroidism and decreased production of thyroid hormones termed as hypothyroidism.

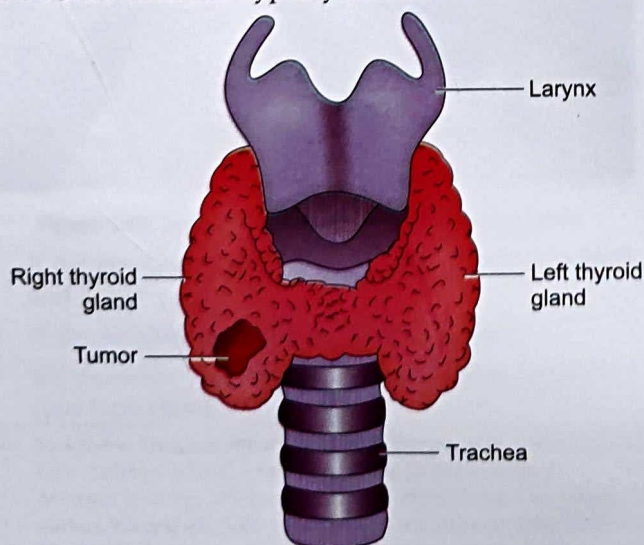


Figure 151: Functional anatomy of thyroid gland

The **secretions secreted by thyroid gland**¹ are T₃ (triiodothyronine – 7% of thyroid hormones), T₄ (thyroxine – 93% of thyroid hormones) and calcitonin. T₃ is the most active form; T₄ is having only minimal hormonal activity. T₄ is much longer lasting and can be converted to T₃ in most tissues and thus T₄ acts as reservoir for T₃. Calcitonin is released in relation to hypercalcemia and lowers the serum calcium levels.

Questions of endocrine dysfunction generally revolve around hyper and hypofunction of specific glands with consequent clinical sequelae. In case of thyroid dysfunction, clinical assessment begins with analysis of categorisation of hyperfunction and hypofunction of thyroid gland. Most thyroid diseases are more common in women than men.

Hyperthyroidism

- ① Clinical features of marked thyroid overactivity are termed as thyrotoxicosis. Milder degrees of hyperthyroidism do not have all the toxic aspects and may only be confirmed after biochemical investigation. The diagnosis is based on the presence of one or two suspicious signs.
- ② Important features include heat intolerance and excessive sweating; increased irritability and emotional lability, palpitations, dyspnoea, tremors, weakness and marked weight loss despite a good appetite, menstrual irregularity most commonly, scanty menstruation, excessive thirst.

1

Thyroid hormones can increase basal metabolic rate for about 60 to 100 percent above normal when large quantities of the hormones are secreted.

- The hands may display a fine tremor which is appreciated by placing a piece of paper on the outstretched hands.
- Two important signs of hyperthyroidism observed are lid retraction and lid lag signs which are appreciated by exophthalmos and Von Graefe's sign respectively. In lid retraction, a rim of white sclera is seen above the iris when the patient is at rest and looks ahead. In assessing lid lag, Von Graefe's sign is performed. In this test, the observer moves his finger slowly downwards from above seated patient, who is asked to watch the finger all the way without moving the head; the upper eyelid is then seen to lag behind the eyeball.

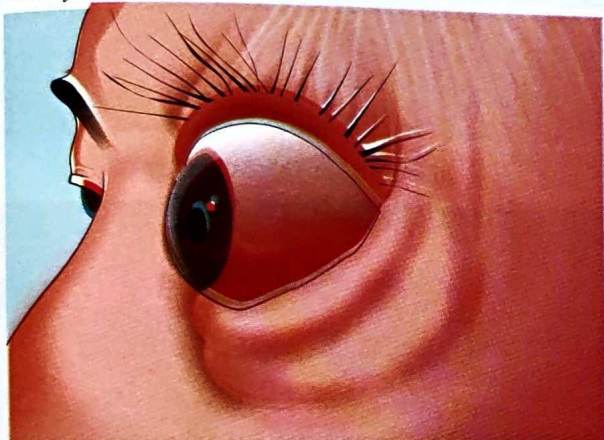


Figure 152: Exophthalmos or Von Graefe's sign



Figure 153: Lid lag seen in patient of hyperthyroid

Hypothyroidism

- In elderly individuals, the patient may have lethargy, loss of concentration and memory. Cold intolerance is the frequent symptom, paraesthesia in the distribution of the median nerve in the hand due to carpal tunnel syndrome.
- Deafness and unsteady gait may also be observed associated with aches and pains in the muscles and joints. Weight gain is also seen. Constipation is also common.
- Hypothyroidism is difficult to diagnose in young

women who are presented with menorrhagia, infertility or recurrent abortion.

- Low IQ (Intelligence Quotient) and short stature are the commonest sequelae observed in hypothyroidism.

Method of physical examination of the patient of thyroid disease

Physical examination in the patient of thyroid dysfunction can be categorised into **general survey**, **local examination** and **general examination**.

(a) General survey

- **Build and state of nutrition of the patient:** In thyrotoxicosis, the patient is usually thin and underweight. There will be excess sweating along with muscle wasting. In hypothyroidism, the patient is obese and overweight. In case of thyroid carcinoma, there will be signs of anaemia and cachexia.
- **Facies:** In thyrotoxicosis, facial expression of excitement, tension, nervousness or agitation with or without exophthalmos is seen. In case of hypothyroidism, one can see the puffiness of the face without any expression.
- **Mental status and intelligence:** Hypothyroid patients are most commonly seen as dull patients.
- **Pulse rate:** In thyrotoxicosis, pulse rate is irregular, whereas in hypothyroidism, there will be bradycardia.
- **Skin:** Skin is moist, particularly of hands in primary thyrotoxicosis. Skin is dry and inelastic in myxoedema.

(b) Local examination

- Local examination of the thyroid gland is performed by examination of the neck where the clinician stands behind the patient and performs the examination. **Inspection**, **palpation** and **auscultation** are performed.

i. Inspection

- **Pizzillo's method:** In this method, the hands are placed behind the head and the patient is asked to push his/her head backwards against the clasped hands on the occipital region of the head.



Figure 154: Pizzillo's method

- Ask the patient to swallow and watch for the movement of thyroid gland. Normally, thyroid gland moves upwards on deglutition.

ii. Palpation

This should be done with the patient's neck slightly flexed. The gland may be palpated from the behind and from the front. The patient should be seated on the stool and the clinician stands behind the patient. Palpation should be carried out in entirety, every part of the gland is palpated to note about the nature of the swelling including size, shape, consistency, mobility of the swelling etc, nodule or hardness of the gland. Palpation of the cervical lymph nodes should be done while considering the diagnosis of thyroid malignancy.

iii. Auscultation

- In primary toxic goitre, a systolic bruit may be heard over the goitre due to increased vascularity.

(c) General examination:

- Eye signs:** 4 types of important changes can be observed in the eyes in case of thyrotoxicosis.

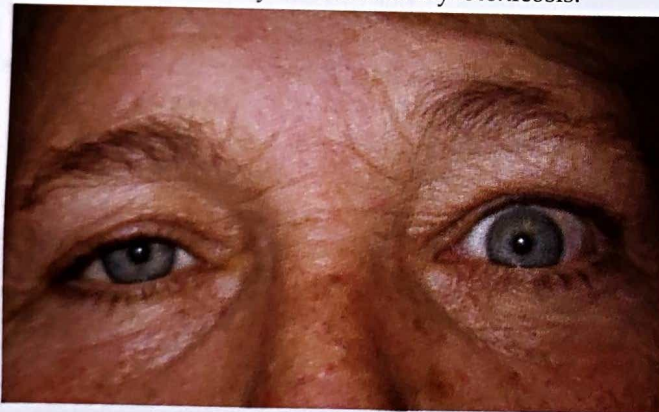


Figure 155: Lid retraction

- Lid retraction:** In this upper eyelid is lying higher than the normal and lower eyelid is in its normal position. (see Figure 155)
 - Exophthalmos:** In this situation, the eyeball is pushed forwards due to the increase in fat or oedema or cellular infiltration in retro orbital space. (see Figure 156)
 - Ophthalmoplegia**
 - Chemosis:** This is oedema of conjunctiva. This may occur due to obstruction to drainage of the conjunctiva by the increased retro orbital pressure. (see Figure 157)
- Tachycardia:** Tachycardia without raise of body temperature can be seen in primary toxic goitre.
 - Tremors of the hands:** fine tremor is present in primary thyrotoxic case. In severe cases, tongue and fingers are together examined which may tremble during examination. (see Figure 158)

- Moist skin:** of the hands and feet are common in case of primary thyrotoxic cases.
- Searching for metastasis:** Skull, spine, ends of long bones, pelvis should be examined for metastasis.



Figure 156: Exophthalmos

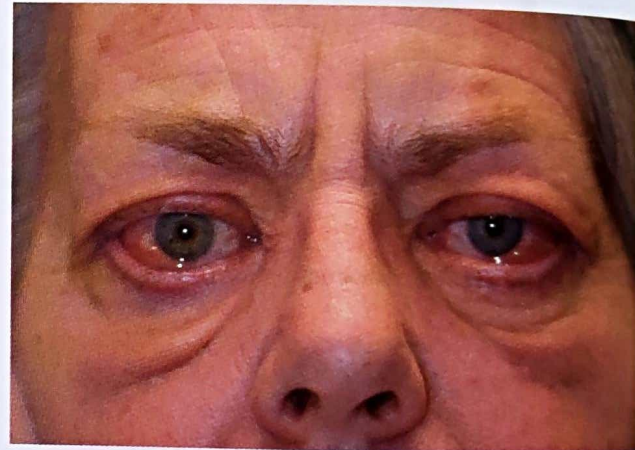


Figure 157: Chemosis

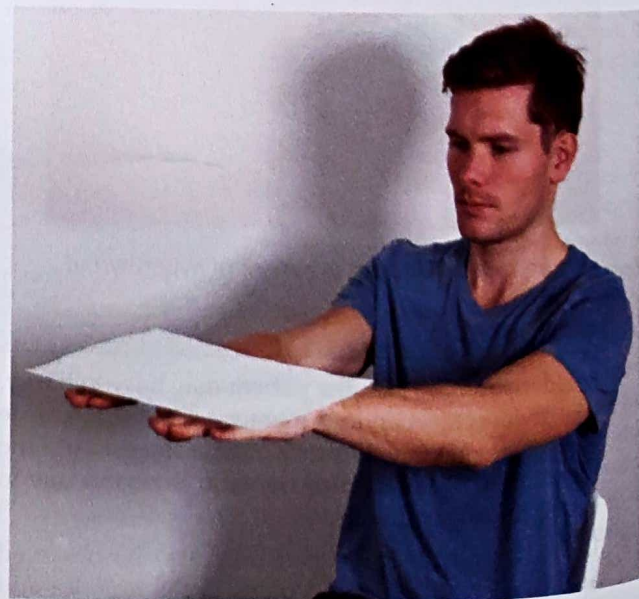


Figure 158: Tremors of hands

Investigations and interpretation

- Thyroid function test (TFT) is performed in the diagnosis of thyroid disease. It includes the analysis of T3, T4 and TSH levels in the blood.
- Thyroid Stimulating Hormone (TSH) measurement is the best means of determining thyroid dysfunction. This is secreted from anterior pituitary gland. Its normal results rules out the diagnosis of hypofunction and hyperfunction of thyroid gland.

Status of thyroid function	Serum TSH	Serum T4	Serum T3
Hyperthyroidism	Low	High	High
T3 toxicosis	Low	Normal	High
Primary hypothyroidism	High	Low	Low or normal
Secondary to pituitary disease	Low or normal	Low	Low or normal

Self-Assessment: Questions for Reviews

Short essay questions

- Explain thyroid dysfunction in brief
- Explain hyperthyroidism
- Explain hypothyroidism
- Explain the clinical features of thyroid dysfunction
- How to do the examination of thyroid gland?

MCQs

- Weight gain is observed in
 - Hypothyroidism;
 - Hyperthyroidism;
 - Both hypothyroidism and hyperthyroidism;
 - None of the above

Answer: (a)

- Von Graefe's sign is helpful in the diagnosis of
 - Hypothyroidism;
 - Hyperthyroidism;
 - Hypopituitarism;
 - Hypogonadism

Answer: (b)

- Which of the following hormone is not secreted by thyroid gland?

- Triiodothyronine;
- Tetraiodothyronine;
- Calcitonin;
- TSH

Answer: (d)

- Goitre is seen in the patients who are having the deficiency of

- Calcium;
- Iodine;
- Sodium;
- Potassium

Answer: (b)

- The following are the features of thyrotoxicosis except

- Exophthalmos;
- Von Graefe's sign;

- Weight loss;
- Weight gain

Answer: (d)

- Congenital thyroid agenesis leads to the manifestation of

- Acromegaly;
- Cushing's syndrome;
- Cretinism;
- Conn's syndrome

Answer: (c)

- Hypothermia can be seen in

- Hyperthyroidism;
- Hyperparathyroidism;
- Myxoedema;
- Goitre

Answer: (c)

- Increased BMR (basal metabolic rate) is seen in which of the following condition?

- Hyperthyroidism;
- Hypertension;
- Acromegaly;
- Starvation

Answer: (a)

- Graves disease is another name of

- Exophthalmic goitre;
- Cretinism;
- Myxoedema;
- Mangolism

Answer: (a)

- Cretinism is caused by the insufficiency of

- Vitamin A;
- Vitamin D;
- Calcium;
- Thyroid hormones

Answer: (d)

Chapter resources

- <https://www.vectorstock.com/royalty-free-vector/thyroid-gland-and-larynx-diagram-vector-38234643>
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
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- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
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